

POWER

POWER - the rate at which energy is transformed, or the rate at which work is done.

$$P = \frac{W}{t} = \frac{E}{t}$$

where P = the power generated (W - Watts)
 W = the work done (J)
 E = the energy transformed (J)
 t = the time taken for the process (s)

Calculating the power developed is straightforward when there is a change in mechanical energy or work, but some situations involve movement at a constant speed which involves no change in mechanical energy.

Using $W = Fs$, the equation changes as follows.

$$P = \frac{W}{t} = \frac{Fs}{t} \\ = F v_{av}$$

where F = average force applied (N)
 v_{av} = the average velocity of the object (ms^{-1})

EXAMPLES

1. An electric motor drives a lift of nett mass 1.47×10^4 kg through a height of 27.0 m in 12.0 s. What is the power of the motor?

2. A cyclist in a road race is producing an average of $5.00 \times 10^2 \text{ W}$ of power whilst moving at 48.0 kmh^{-1} along a flat road. Ignoring any air resistance, what average force is being supplied through the pedals?